

ORIGINAL ARTICLE

Single Interventions (Pharmacological or Renal Denervation) Are Not Sufficient to Achieve Blood Pressure Control in Resistant Hypertension a Systematic Review and Meta-Analysis

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BACKGROUND: Resistant hypertension remains a major clinical challenge. This systematic review/meta-analysis evaluated whether the addition of a single pharmacological or device-based intervention (renal denervation [RDN]) can achieve effective blood pressure (BP) control in patients with true resistant hypertension.

METHODS: A systematic search of MEDLINE/PubMed was performed to identify studies assessing the antihypertensive effects of adding a single pharmacological or device-based intervention in patients with true resistant hypertension. Primary analyses were conducted using a single-arm framework.

RESULTS: Sixty-eight studies were included (n=6297; weighted mean age, 60 years; males 60%; diabetes 35%; smoking 16%; cardiovascular disease 27%). Participants received an average of 4.8 antihypertensive medications, with baseline office BP of 164/92 mmHg and 24-hour ambulatory BP of 148/85 mmHg. The median follow-up was 6 months. Meta-analysis of 58 studies (n=4579; 52% RDN) demonstrated a pooled mean 24-hour systolic ambulatory BP reduction from baseline of -11.3 mmHg (95% CI, -12.3 to -10.2). Meta-analysis of 15 studies (n=2700; 15% RDN) showed a pooled hypertension control rate during follow-up of 35% (95% CI, 28–42). No significant differences were observed between RDN and pharmacotherapy or between randomized and nonrandomized studies.

CONCLUSIONS: A single intervention—either pharmacological or RDN—on top of guideline-directed background therapy resulted in clinically meaningful BP reduction in patients with true resistant hypertension; however, only one-third of patients achieved BP control. Future clinical trials are needed to evaluate whether combination treatment strategies integrating optimized pharmacological regimens with RDN can provide more effective and durable BP control in this particularly challenging patient population. (*Hypertension*. 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.126.26756.) • **Supplement Material.**

Key Words: blood pressure ■ classification ■ denervation ■ humans ■ smoking

"What cannot be healed by medicine is healed by iron. What iron cannot heal is healed by fire. And what fire cannot heal, one must consider incurable."
Hippocrates (460 BC)

Resistant hypertension (RHTN), which accounts for 5% to 10% of treated hypertensive patients, remains a major clinical challenge. What should be the next step after the established triple combination

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NOVELTY AND RELEVANCE

What Is New?

The addition of any antihypertensive intervention—whether pharmacological or device-based—in patients with true resistant hypertension (RHTN) receiving guideline-directed background therapy produced a meaningful blood pressure (BP)-lowering effect. However, only about one-third of these patients achieved adequate BP control.

What Is Relevant?

RHTN remains a significant challenge in clinical practice. This systematic review and meta-analysis evaluated the BP-lowering efficacy and hypertension control rates achieved with pharmacological or

device-based interventions in patients with RHTN. Based on these findings, a comprehensive management algorithm for RHTN is proposed, and this strategy should be assessed in future clinical trials.

Clinical/Pathophysiological Implications?

A single intervention (pharmacological or renal denervation) on top of guideline-directed background therapy with a triple single-pill combination is not sufficient to achieve BP control in most patients with RHTN. Combination treatment strategies that integrate optimized pharmacological regimens beyond the standard triple single-pill combination with device-based interventions may offer more effective and durable BP control in this particularly challenging patient population.

Nonstandard Abbreviations and Acronyms

ABP	ambulatory blood pressure
BP	blood pressure
RDN	renal denervation
RHTN	resistant hypertension
RoB	risk of bias
SPC	single-pill combination

therapy is debatable. Most guidelines recommend adding a mineralocorticoid receptor antagonist, such as spironolactone, or eplerenone when the former is contraindicated or not tolerated, as the most effective therapeutic option.^{1–4} Additional agents, including β -blockers, α -blockers, or centrally acting sympatholytics, may also be considered to achieve further blood pressure (BP) reduction.^{1–4} Renal denervation (RDN) has recently been incorporated into hypertension management algorithms, with true RHTN being one of its key indications.^{1–4} The procedure has demonstrated both safety and efficacy, providing durable BP reductions that are independent of patient adherence due to its always-on mechanism of action.

Given these developments, a key question remains: is the currently recommended stepwise treatment algorithm sufficient to achieve optimal BP control in patients with true RHTN? To address this, the aim of the present systematic review and meta-analysis was to synthesize data from relevant studies and evaluate: (1) the BP-lowering effects of pharmacological interventions (including novel antihypertensive agents) or RDN in patients with true RHTN; (2) the proportion of patients who ultimately achieve BP control following the addition

of pharmacological therapy or RDN; and (3) based on these findings, propose a pragmatic treatment algorithm for the effective and durable management of true RHTN that should be evaluated in future clinical trials.

METHODS
Data Sources and Search Strategy

The data that support the findings of this study are available from the corresponding author upon reasonable request. A systematic literature search was conducted in the MEDLINE/PubMed database through September 9, 2025, in line with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses recommendations⁵ using the following search algorithm: resistant hypertension. Potentially eligible articles were also hand-searched from the reference lists of relevant articles according to the snowball approach. The literature search was performed independently by 3 investigators, and disagreements were resolved by consensus with senior authors.

Study Selection

Studies were considered eligible for inclusion if they: (1) included patients with true RHTN; and (2) evaluated the antihypertensive effect of the addition of a single pharmacological or device-based intervention (studies were eligible either if they provided pre- and postintervention BP measurements or pre- versus postintervention BP differences). Patients in the included studies were required to have uncontrolled RHTN, defined according to recent guidelines as follows: office BP $\geq 140/90$ mmHg on at least 3 antihypertensive drugs at maximum recommended and tolerated doses, including a diuretic; uncontrolled status confirmed with out-of-office BP (ambulatory BP [ABP] or home BP) monitoring; and secondary hypertension excluded.^{1–3} Both observational and randomized, single- or multiarm studies were included. Studies with overlapping populations, case reports, cross-sectional studies,

editorials, narrative reviews, and meta-analyses were excluded. In addition, studies with apparent RHTN or mixed (uncontrolled and RHTN) populations, employing external or surgical RDN, or using indirect mean BP difference estimates (eg, least square mean BP changes) were excluded.

Data Extraction

Three investigators performed the extraction and tabulation of the data from each study. Extracted data included the basic characteristics of each study (author, year, country, design, RHTN definition, intervention, follow-up), baseline characteristics of included participants (setting [general RHTN population or special RHTN subgroup], demographics, comorbidities), details regarding the BP measurement methodology (devices used, number/d of measurements, day-night periods, and measurement intervals), details regarding the interventions employed (catheter systems and technologies, eg, radiofrequency or ultrasound, drug dosages), and data regarding the BP (in mmHg units) of participants after versus before the implementation of each intervention.

Risk of Bias Assessment

The risk of bias (RoB) of the included studies was independently assessed by 3 investigators using the National Heart, Lung, and Blood Institute/National Institutes of Health Quality Assessment Tool for Before-After (Pre-Post) Studies With No Control Group.⁶ The Question 12 of the tool was considered irrelevant and was excluded; thus, an 11-item tool was used to assess the methodological quality of the included studies in addressing the research question of this systematic review and meta-analysis. Points were assigned to studies depending on the fulfillment or not of each item (Yes: 1 point; Unclear/Not reported: 0.5 points; No: 0 points), with a maximum score of 11. The RoB categories were defined as follows: (1) high RoB (0.0–3.5 points); (2) moderate RoB (4.0–7.5 points); and (3) low RoB (8.0–11.0 points). A detailed illustration regarding the assessment of the RoB is provided in the [Supplemental Material](#).

Statistical Analysis

A meta-analysis was conducted to calculate the pooled mean BP difference (weighted mean difference) after versus before the implementation of the intervention in each study (ie, the mean change in BP [in mmHg units] from baseline to follow-up). The primary BP component of interest was the 24-hour ABP and a meta-analysis stratifying for the type of intervention used (pharmacotherapy or RDN) was conducted. In case the SD of the mean after-before difference was not provided, it was calculated using appropriate formulas for paired analyses ($SD_{dif} = \sqrt{SD_{before}^2 + SD_{after}^2 - 2 \times SD_{before} \times SD_{after} \times r}$), assuming a Pearson correlation coefficient (r) of 0.5, a commonly used mid-range value for physiological repeated measures such as BP (this calculation was required for 21 reports included in the primary end point meta-analysis).⁷ This value provides a conservative estimate of the within-subject correlation, while sensitivity analyses (not presented) using alternative values ($r=0.25$ and 0.75) yielded similar results. In case a study provided data for multiple follow-up time points, the most recent and informative was used so that treatment modifications

other than those under investigation were kept to a minimum. Most of the included studies were single-arm studies; thus, for the primary analysis, all studies irrespective of their primary design were analyzed as single-arm studies. For example, in case of a placebo-randomized controlled trials, the within-group BP change of the intervention arm was taken into consideration and in case of an randomized controlled trials with 2 treatment arms, the within-group BP change of each arm was analyzed separately. This separate consideration of the multiple arms of a single study contributed to avoidance of double-counting patients when multiple reports were derived from the same study. Sensitivity analyses were performed to assess the impact of antihypertensive interventions on different BP measurement methods (office BP, home BP, daytime and nighttime ABP), considering the primary design of each study (eg, analysis with placebo-controlled BP differences in case of placebo-controlled randomized controlled trials), and dividing studies according to their median sample size ($\geq$ versus <50 participants), their median follow-up ($\geq$ versus <6 months), and the origin/geographic region of their participating centers. Influence leave-one-out analyses were also conducted to assess the influence of individual studies on the overall pooled estimates. Meta-regression analyses were employed to assess the association of pooled estimates with baseline characteristics of the included studies and their populations (age, sex, body mass index, diabetes prevalence, smoking prevalence, cardiovascular disease prevalence, baseline BP, estimated glomerular filtration rate, number of drugs at baseline, follow-up period, overall RoB score, and year of publication). A meta-analysis of proportions was finally conducted to assess the control status of participants after the implementation of the intervention in each study.

Heterogeneity was evaluated using the I^2 statistic, and a random-effects model was used for all analyses. Comparison of pooled treatment effects between the RDN and pharmacotherapy groups was performed using the χ^2 test for subgroup differences (Cochran Q test for heterogeneity between subgroups). Small-study effects and potential publication bias were assessed by inspection of funnel plots, trim-and-fill analyses, and by using formal statistical Egger and Begg tests. Baseline characteristics of participants in the included studies were summarized as weighted means for continuous variables and percentages for categorical variables. Appropriate formulas were used to provide aggregate study baseline data in case characteristics were given for subgroups⁸ or as median (interquartile range) instead of mean $\pm$ SD.⁹ Analyses were performed using the R Statistical Software (v4.4.2; R Core Team 2024), and a 2-sided $P<0.05$ was considered as statistically significant.

RESULTS

Search Results and Study Characteristics

After the literature search, 4501 records were initially retrieved, among which 68 were deemed eligible for inclusion.^{10–77} A summary of the main characteristics of each study is shown in Table. Most studies were conducted after 2010 in various countries across the globe and included general RHTN populations, with only a few recruiting special population subgroups (eg, chronic

Table. Main Baseline Characteristics of the Included Studies (Mean for Continuous and % for Categorical Variables)

Study	Setting Country	Intervention	Follow-up, mo	N	Age, y	Males, %	BMI, kg/m ²	Diabetes, %	Smoking, %	CVD, %	eGFR, mL/min per 1.73 m ²	Office SBP/DBP, mm Hg	24-hour SBP/DBP, mm Hg	Number of drugs
Browers 2024 ¹⁰	RHTN Italy, Belgium	RF-RDN	1	65	60	58	28.9	29	20	NR	NR	169/87	153/88	4.34
Al Ghorani 2024 ¹¹	RHTN Germany	RF-RDN	12	39	62	64	32.0	44	8	20	68.7	164/91	153/85	4.90
Wang 2024 ¹²	RHTN China	RF-RDN	12	60	50	57	26.0	32	55	40	78.4	179/101	150/92	3.65
Püşüröğlu 2023 ¹³	RHTN Turkey	Spironolactone Nebivolol	2	81	58	51	31.9	31	28	NR	NR	147/82	158/96	NR
Manukyan 2022 ¹⁴	RHTN Russia	RF-RDN	6	59	60	42	35.2	100	NR	60	69.7	171/88	158/83	4.3
Galceran 2023 ¹⁵	RHTN Spain	Spironolactone	12	58	67	64	31.6	48	NR	10	NR	152/83	147/77	3.8
Schlaich 2022 (PRECISION) ¹⁶	RHTN International	Aprocitan	1	486	61	60	34.0	55	NR	NR	NR	153/88	138/83	NR
Mahfoud 2021 ¹⁷	RHTN International	Alc-RDN	3	45	55	62	30.7	33	NR	NR	85.0	169/99	151/89	5.1
Azizi 2021 (RADIANCE-HTN TRIO) ¹⁸	RHTN International	US-RDN	2	69	52	81	32.8	30	NR	10	86.0	156/101	144/89	4.0
Kim 2021 ¹⁹	RHTN South Korea	RF-RDN	3	16	57	75	27.9	31	6	19	77.7	164/102	151/91	4.0
Ranasinghe 2020 (APPROPRIATE) ²⁰	RHTN Sri Lanka	Propanolol	3	18	56	28	28.4	NR	NR	NR	NR	NR/ NR	149/85	NR
Scalise 2020 ²¹	RHTN+HD Italy	RF-RDN	6	12	57	67	NR	33	NR	NR	HD	180/100	175/96	6.0
Rodriguez-Leor 2020 (Flex-Spyral Registry) ²²	RHTN Spain	RF-RDN	6	68	NR	NR	NR	NR	NR	NR	NR	NR	151/89	4.9
Ukena 2020 ²³	RHTN Germany	RF-RDN	6	105	64	67	NR	47	NR	27	NR	171/92	148/84	5.3
Daemen 2019 (ACHIEVE) ²⁴	RHTN Germany, the Netherlands, Sweden	US-RDN	12	96	64	59	30.1	40	9	NR	91.9	176/95	156/88	5.1
Fengler 2019 (RADIOSOUND-HTN) ²⁵	RHTN Germany	RF/US-RDN	3	120	64	69	31.6	46	46	36	77.4	NR/ NR	150/84	4.7
Krieger 2018 (ReHOT) ²⁶	RHTN Brazil	Spironolactone Clonidine	3	187	55	46	30.7	33	9	10	88.9	154/92	143/86	NR
Ott 2018 ²⁷	RHTN Germany	RF-RDN	3	94	65	64	30.1	46	NR	NR	66.8	170/92	153/89	5.5
Worthley 2017 ²⁸	RHTN Australia, New Zealand	RF-RDN	3	39	63	62	NR	33	NR	15	74.0	174/93	156/83	NR
Feyz 2017 ²⁹	RHTN the Netherlands	RF/US-RDN	6	31	64	48	29.0	26	36	NR	NR	182/94	150/83	NR
Jacobs 2017 (INSPIRED) ³⁰	RHTN Belgium	RF-RDN	6	6	48	50	29.3	0	66	NR	93.5	173/107	161/101	4.0
Daniels 2017 ³¹	RHTN+OSA Ireland	RF-RDN	6	20	55	85	37.2	65	NR	NR	NR	166/93	154/87	5.5
Kalizki 2017 ³²	RHTN Germany	Eplerenone	6	25	62	76	29.3	0	NR	NR	NR	166/91	143/82	3.8
Pekarskiy 2017 ³³	RHTN Russia	RF-RDN	6	51	56	39	NR	41	NR	47	78.0	170/96	164/91	4.2
Tsioufis 2017 ³⁴	RHTN Greece	RF-RDN	6	79	59	70	33.0	35	14	NR	87.0	176/95	155/90	4.4
Oliveras 2016 (DENERVHTA) ³⁵	RHTN Spain	RF-RDN Spironolactone	6	24	64	63	32.0	50	38	21	80.2	170/90	153/81	4.1
Aripov 2016 ³⁶	RHTN Kazakhstan	RF-RDN	12	63	54	60	30.8	30	NR	10	100.2	186/111	NR/ NR	4.3

(Continued)

Table. Continued

Study	Setting Country	Intervention	Follow-up, mo	N	Age, y	Males, %	BMI, kg/m ²	Diabetes, %	Smoking, %	CVD, %	eGFR, mL/min per 1.73 m ²	Office SBP/DBP, mm Hg	24-hour SBP/DBP, mm Hg	Number of drugs
Mathiassen 2016 (ReSET) ⁷⁶	RHTN Denmark	RF-RDN	1	36	54	75	28.2	25	19	6	NR	160/95	152/91	4.1
Chen 2016 ³⁷	RHTN Kazakhstan	RF-RDN	6	47	49	64	27.2	15	NR	11	85.8	181/104	160/95	4.5
Prochnau 2016 ³⁸	RHTN Germany	RF-RDN	3	70	64	60	31.1	50	NR	20	69.4	NR/NR	165/89	5.9
Kadziela 2016 (RDN-POL Registry) ³⁹	RHTN Poland	RF-RDN	6	44	52	52	NR	27	NR	NR	83.1	174/100	154/90	5.7
Skowerski 2016 ⁴⁰	RHTN Poland	RF-RDN	6	15	62	46	30.9	33	NR	33	77.5	204/108	152/87	5.1
Djoumessi 2016 ⁴¹	RHTN+DM Cameroon	Spironolactone	1	9	65	33	30.3	100	0	NR	55.3	158/86	140/78*	NR
de Sousa Almeida 2016 ⁴²	RHTN Portugal	RF-RDN	12	31	65	48	31.8	77	3	32	76.4	176/90	150/83	5.8
Sharp 2016 ⁴³	RHTN UK	RF-RDN	11	253	57	47	32.0	27	NR	NR	69.0	185/102	NR/NR	5.0
Donazzan 2015 ⁴⁴	RHTN Germany	RF-RDN	6	11	66	55	32.2	45	NR	NR	62.7	180/97	149/85	6.0
Ewen 2015 ⁴⁵	RHTN Germany, Italy	RF-RDN	6	84	65	73	31.0	37	NR	20	67.0	167/88	148/84	5.0
Baroni 2015 ⁴⁶	RHTN Italy	RF-RDN	3	12	69	75	NR	NR	NR	NR	75.6	170/86	NR/NR	6.1
van Brussel 2015 ⁴⁷	RHTN the Netherlands	RF-RDN	1.5	21	59	71	27.9	33	NR	48	73.0	174/100	162/93	4.7
Williams 2015 (PATHWAY-2) ⁴⁸	RHTN UK	Spironolactone Doxazosin Bisoprolol	3	335	61	69	NR	14	7	NR	NR	157/90	148/84*	NR
Mirowska 2015 (ReShape CV-risk) ⁴⁹	RHTN Norway	RF-RDN	6	23	53	78	32.0	0	13	8	83.9	162/108	154/97	4.8
Tuohi 2015 ⁵⁰	RHTN Ireland	RF-RDN	9	23	57	70	NR	NR	NR	NR	90.0	168/94	150/87	4.9
Kario 2015 (SYMPPLICITY-HTN Japan) ⁵¹	RHTN Japan	RF-RDN	6	22	60	68	27.0	36	32	NR	NR	181/NR	165/NR	4.9
McLellan 2015 ⁵²	RHTN Australia	RF-RDN	6	14	64	67	31.0	14	NR	NR	75.0	163/87	152/84	4.9
Azizi 2015 (DENERHTN) ⁵³	RHTN France	RF-RDN	6	53	55	64	30.7	17	57	30	NR	159/93	152/90	NR
Ziegler 2015 ⁵⁴	RHTN Germany	RF-RDN	6	24	78	50	NR	8	NR	41	64.0	173/86	158/80	4.3
Verheye 2014 (RAPID) ⁵⁵	RHTN Europe, New Zealand	RF-RDN	6	50	63	58	32.6	42	44	26	75.8	182/96	154/88	5.1
Sievert 2014 (REDUCE-HTN) ⁵⁶	RHTN Europe, Australia, New Zealand	RF-RDN	6	146	59	61	NR	28	NR	NR	83.9	182/100	153/88	5.3
Rosa 2014 (Prague-15) ⁵⁷	RHTN Czech Republic	RF-RDN Spironolactone	6	106	56	70	32.4	20	15	7	NR	157/91	148/85	5.2
Verloop 2014 ⁵⁸	RHTN the Netherlands	RF-RDN	12	54	58	50	29.2	15	NR	NR	76.0	197/108	162/98	6.1
Ni 2014 ⁷⁷	RHTN+HD China	Spironolactone	3	40	56	60	NR	NR	NR	NR	NR	154/93	147/91	5.1
Luetkens 2014 ⁵⁹	RHTN Germany	RF-RDN	9.5	17	55	76	28.6	NR	NR	NR	73.8	177/104	155/94	4.7

(Continued)

Table. Continued

Study	Setting Country	Intervention	Follow-up, mo	N	Age, y	Males, %	BMI, kg/m ²	Diabetes, %	Smoking, %	CVD, %	eGFR, mL/min per 1.73 m ²	Office SBP/DBP, mm Hg	24-hour SBP/DBP, mm Hg	Number of drugs
Dong 2014 ⁶⁰	RHTN China	RF-RDN	3	5	46	60	NR	40	NR	NR	82.4	166/107	161/104	NR
Bakris 2014 (SYMPPLICITY HTN-3 ambulatory) ⁶¹	RHTN Europe, Australia, New Zealand	RF-RDN	6	364	58	59	34.2	47	NR	NR	NR	NR/NR	159/88	NR
Bhatt 2014 (SYMPPLICITY HTN-3 office) ⁶²	RHTN Europe, Australia, New Zealand	RF-RDN	6	364	58	59	34.2	47	NR	NR	NR	NR/NR	NR/NR	NR
Fadl Elmula 2014 ⁶³	RHTN Norway	RF-RDN	6	9	57	78	29.0	22	NR	11	NR	156/91	152/93	5.1
Volz 2014 ⁶⁴	RHTN Sweden	RF-RDN	6	22	61	50	30.0	23	NR	18	98.0	169/96	158/88	4.0
Kiuchi 2013 ⁶⁵	RHTN+CKD Brazil	RF-RDN	6	24	55	38	32.0	38	NR	NR	64.4	186/108	151/92	4.7
Damascelli 2013 ⁶⁶	RHTN Italy	RF-RDN	6	24	66	79	NR	46	NR	NR	69.8	185/105	162/91	NR
Kaltenbach 2012 ⁶⁷	RHTN Germany	RF-RDN	6	20	61	55	NR	50	NR	NR	77.4	149/83	147/80	NR
Witkowski 2011 ⁶⁸	RHTN+OSA Poland	RF-RDN	6	10	50	70	30.6	40	NR	NR	82.1	173/104	138/84	4.8
Václavík 2011 (ASPIRANT) ⁶⁹	RHTN Czech Republic	Spironolactone	2	55	61	67	32.3	NR	NR	NR	NR	156/93	143/81	NR
Bakris 2010 (DORADO-AC) ⁷⁰	RHTN International	Darusentan Guanfacine	3.5	729	62	45	32.0	41	NR	23	81	152/88	134/78	NR
Böhm 2010 (SYMPPLICITY HTN-2) ⁷¹	RHTN International	RF-RDN	6	52	58	65	31.0	40	NR	NR	77.0	178/97	NR/NR	5.2
de Souza 2010 ⁷²	RHTN Brazil	Spironolactone	7	175	62	27	30.2	33	10	52	NR	169/92	149/86	NR
Rodilla 2009 ⁷³	RHTN Spain	Spironolactone Doxazosin	3 6	181	65	71	32.5	52	9	NR	76.1	165/85	145/82	NR
Martin 2005 ⁷⁴	RHTN Ireland	Moxonidine	1.5	14	72	57	NR	NR	NR	NR	NR	196/104	163/87	2.6
Sermiswan 2003 ⁷⁵	RHTN Thailand	Methyldopa	1.5	23	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR

Alc indicates alcohol; BMI, body mass index; CKD, chronic kidney disease; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; HD, hemodialysis; NR, not reported; OSA, obstructive sleep apnea; RDN, renal denervation; RF, radiofrequency; RHTN, resistant hypertension; and US, ultrasound.

*Home instead of ambulatory blood pressure.

kidney disease or hemodialysis patients). A total of 6297 participants with true RHTN were included, with the following summary characteristics: weighted mean age 60 years, males 60%, body mass index 32.0 kg/m², diabetes 35%, smoking 16%, history of cardiovascular disease 27%, estimated glomerular filtration rate 79.7 mL/min per 1.73 m², number of antihypertensive drugs 4.8, office BP (systolic/diastolic) 164/92 mm Hg, 24-hour ABP 148/85 mm Hg. The median follow-up period of the included studies was 6 months. Nine studies provided 2 reports^{13,16,26,33,35,37,57,70,73} and 1 study provided 3 reports,⁴⁸ resulting in a total of 79 reports.

The primary design of the included studies was mainly observational, with one-third of the studies being randomized controlled trial, (either placebo/sham- or active-controlled; Table S1). The definition of RHTN for each

study is shown in Table S1. Office BP was measured using validated oscillometric upper-arm cuff BP monitors in the majority of studies (not clear in most studies if attended or unattended), ambulatory BP was the out-of-office method most commonly employed and a detailed summary of the different BP measurement methodologies used in each study is shown in Table S2. The intervention that was investigated in 71% of the studies (50% of the total sample) was RDN (n=3144 participants), namely radiofrequency-RDN (Table S3). The definition of hypertension control among the different studies is shown in Table S4.

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 Checklist for the present systematic review and meta-analysis is presented in Table S5. The Preferred Reporting Items for Systematic

Reviews and Meta-Analyses 2020 for Abstracts Checklist is presented in Table S6. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 flow diagram for systematic reviews and meta-analyses is presented in Figure 1.

BP Reduction

Meta-analysis of 58 studies (67 reports, $n=4579$ participants, 52% RDN) indicated a pooled mean 24-hour systolic ABP difference of -11.3 mm Hg (95% CI, -12.3 to -10.2 ; Table S7; Figures 2 and 3). No significant difference in BP reduction was observed between studies using RDN versus pharmacotherapy ($P=0.92$). The respective funnel plot and the formal statistical tests revealed the potential presence of small-study effect/publication bias; however, without evidence of a substantial effect, as shown in the trim-and-fill analysis (Figure S1). No signals of influence by individual studies were raised in the leave-one-out analysis (Figure S1). Most studies were deemed as of low RoB (Table S8). The respective forest plot for the pooled mean 24-hour diastolic ABP difference (52 studies, 60 reports, $n=3533$ participants, 52% RDN) is shown in Figure 2 and Figure S2. Meta-analysis of placebo/sham-controlled randomized studies showed a pooled mean 24-hour systolic ABP difference of -9.6 mm Hg (-11.1 to -8.1) when analyzed as single-arm and a difference of -6.7 mm Hg (-9.0 to -4.3) when placebo/sham-corrected analysis was performed (Figure 4 and Table S7).

In meta-regression analysis, the pooled mean 24-hour systolic ABP treatment-induced decline was positively associated with baseline 24-hour systolic ABP ($P<0.01$) and inversely with the baseline number of drugs ($P<0.01$; Figure S3). In sensitivity analyses, the impact of study interventions on different BP measurement methods (systolic/diastolic office BP, home BP, daytime and nighttime ABP, as provided by the included studies; Table S9) was assessed, and the respective forest plots are shown in Figure 2 and Figures S4 through S8. Meta-analysis of nonrandomized studies showed a pooled mean 24-hour systolic ABP difference of -11.8 mm Hg (-13.2 to -10.4), with slightly better performance of pharmacotherapy (Table S7 and Figure S9). Meta-analysis of randomized studies (either placebo/sham- or active-controlled analyzed as single-arm studies) showed a pooled mean 24-hour systolic ABP difference of -10.5 mm Hg (-12.1 to -9.0), with no significant difference between RDN and pharmacotherapy (Table S7 and Figure S10). Studies with ≥ 50 participants showed a greater pooled reduction in 24-hour systolic ABP than studies with < 50 participants (-12.7 versus -10.0 mm Hg, $P=0.01$; Figure S11). No difference was found between studies with follow-up < 6 months (-10.3 mm Hg) versus those with follow-up ≥ 6 months (-11.8 mm Hg; $P=0.16$; Figure S12). Meta-analysis according to the origin/geographic region of the studies showed heterogeneous findings, with a higher pooled mean 24-hour systolic ABP difference in studies conducted in Asia and a lower difference in studies conducted in

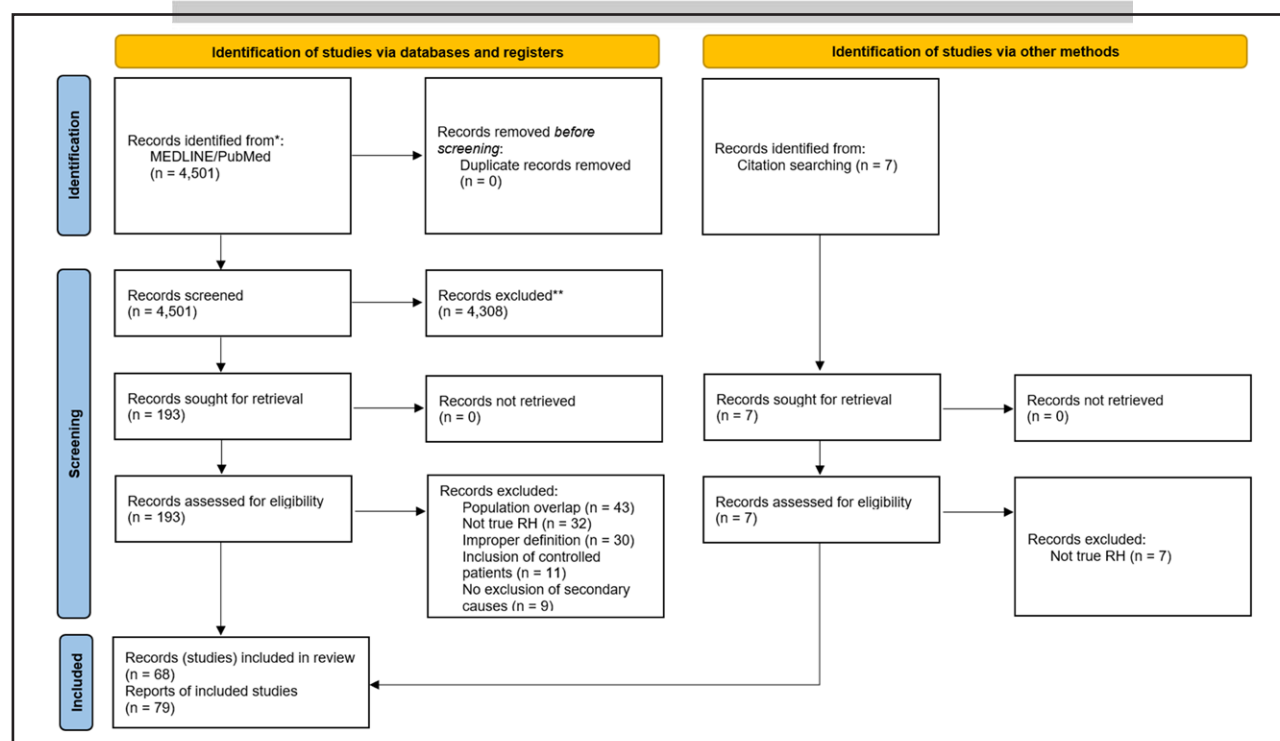


Figure 1. Flowchart for the selection of included studies.

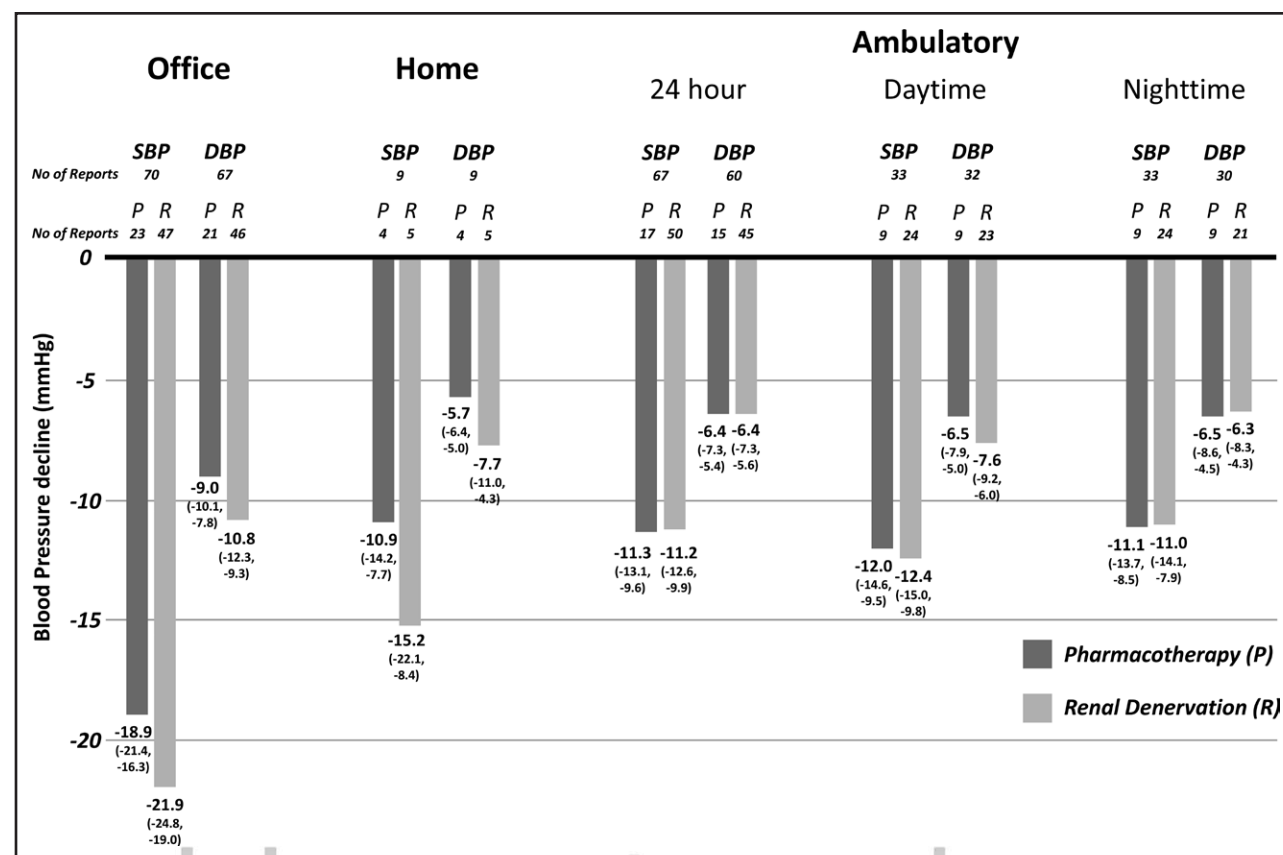


Figure 2. Average systolic blood pressure (SBP)/diastolic blood pressure (DBP) decline in office, home, and ambulatory blood pressure after the addition of pharmacotherapy or renal denervation in patients with true resistant hypertension (all pharmacotherapy vs renal denervation comparisons are nonsignificant; assuming a single-arm design for the included studies).

the context of multicenter international collaborations (−14.2 versus −8.0 mm Hg, $P<0.01$; Figure S13). Meta-analysis of pharmacotherapy studies divided by whether they used spironolactone or not indicated greater efficacy with spironolactone (−13.1 versus −8.5 mm Hg, $P<0.01$; Figure S14).

Hypertension Control Rate

Meta-analysis of 15 studies (20 reports, $n=2700$ participants, 15% RDN) indicated a pooled hypertension control rate after the intervention employed in each study of 35% (95% CI, 28–42; Figure 5). No significant difference was observed in RDN (33% [24–44]) versus pharmacotherapy (36% [27–45]) studies ($P=0.75$). The respective funnel plot, trim-and-fill analysis, and the formal statistical tests did not reveal any potential presence of small-study effect/publication bias (Figure S15). No concerns were raised in the leave-one-out analysis (Figure S15). Most studies were deemed as of low RoB (Table S8).

In meta-regression analysis, the pooled hypertension control rate was inversely associated with the percentage of cardiovascular disease ($P<0.01$) and the total RoB score ($P=0.02$; Figure S16). No difference was observed

in meta-analysis of nonrandomized versus randomized studies (Figure S17). Meta-analysis of pharmacotherapy studies divided by whether they used spironolactone or not indicated a higher hypertension control rate with spironolactone (0.41 versus 0.31, $P<0.01$; Figure S18).

DISCUSSION

This systematic review and meta-analysis demonstrated that, among patients with true RHTN, the addition of a single antihypertensive intervention—either pharmacological therapy or RDN—was associated with several key findings. First, approximately one-third of patients achieved BP control after a single intervention, with no difference between RDN and pharmacotherapy. There was a mean reduction of $\approx 11/6$ mm Hg in 24-hour systolic/diastolic ABP, with no significant difference between RDN and pharmacotherapy studies. Second, similar BP reductions were observed across other measurement modalities, except for office BP, where greater reductions were reported in RDN studies. Third, BP reductions did not differ significantly between randomized and nonrandomized studies, although smaller effects were noted in placebo- or sham-controlled analyses. Fourth, the magnitude of BP reduction was positively associated with baseline

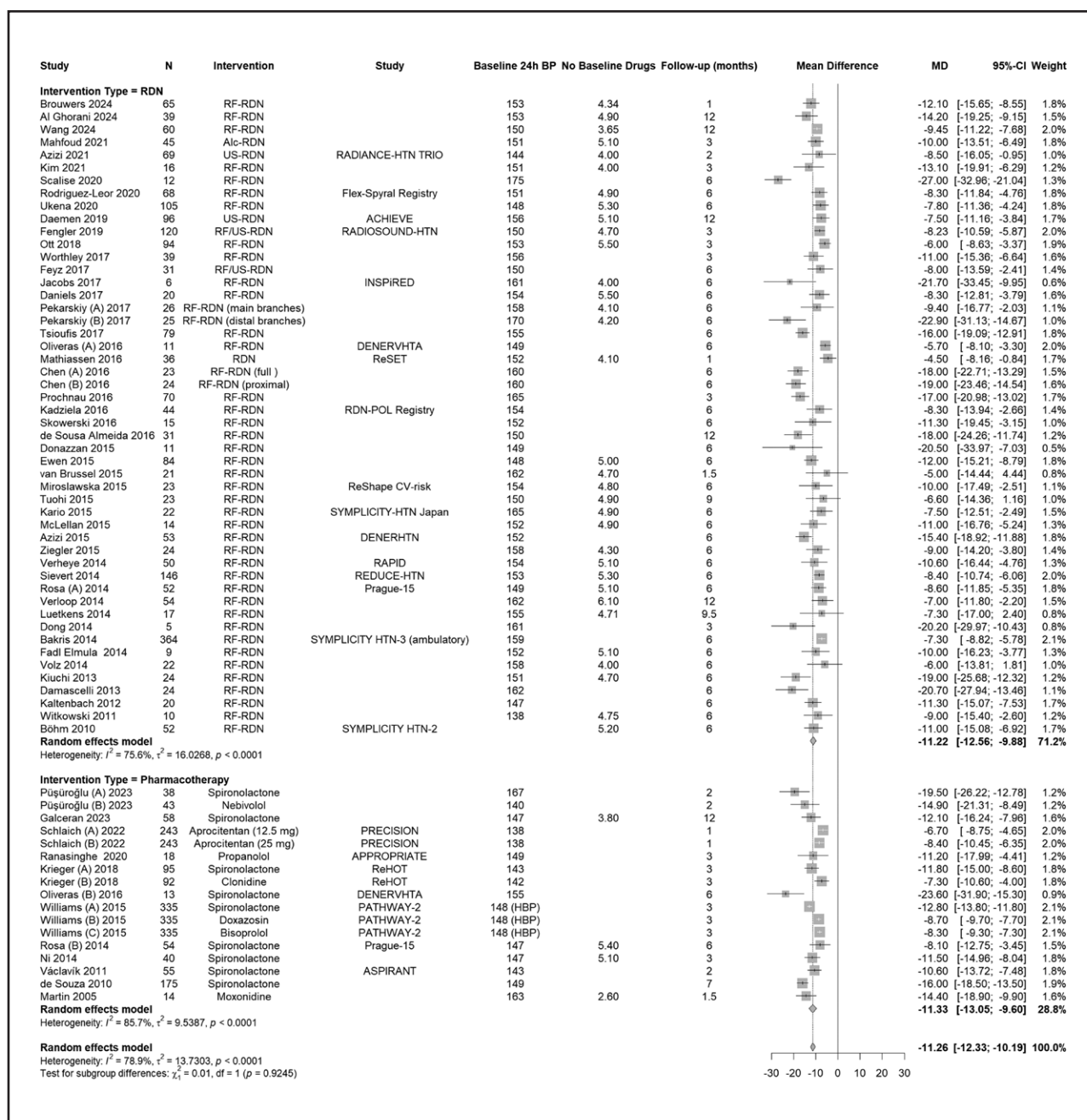


Figure 3. Meta-analysis of mean 24-hour ambulatory systolic blood pressure (BP) difference.

The forest plot is a stratified analysis to compare BP reduction in pharmacotherapy vs renal denervation studies (nonsignificant difference). RDN indicates renal denervation; and RF, radiofrequency.

BP levels and inversely associated with the number of antihypertensive medications used. Fifth, hypertension control rates were comparable between randomized and nonrandomized studies but were influenced by baseline cardiovascular disease and overall RoB scores. Finally, most included studies were considered to have low RoB, and there was no evidence of significant small-study effect or publication bias.

Based on the current definition of true RHTN, only studies that used both office and out-of-office BP

measurements were included. RDN studies more frequently employed ABP monitoring, whereas pharmacotherapy studies more commonly used home BP monitoring. Although BP measurement protocols varied considerably across studies, most clearly described their procedures and appropriately excluded secondary hypertension. The majority of RDN studies evaluated radiofrequency-RDN, with fewer investigating ultrasound-based (ultrasound-RDN) or alcohol-based approaches. Overall, 47% of participants received pharmacological

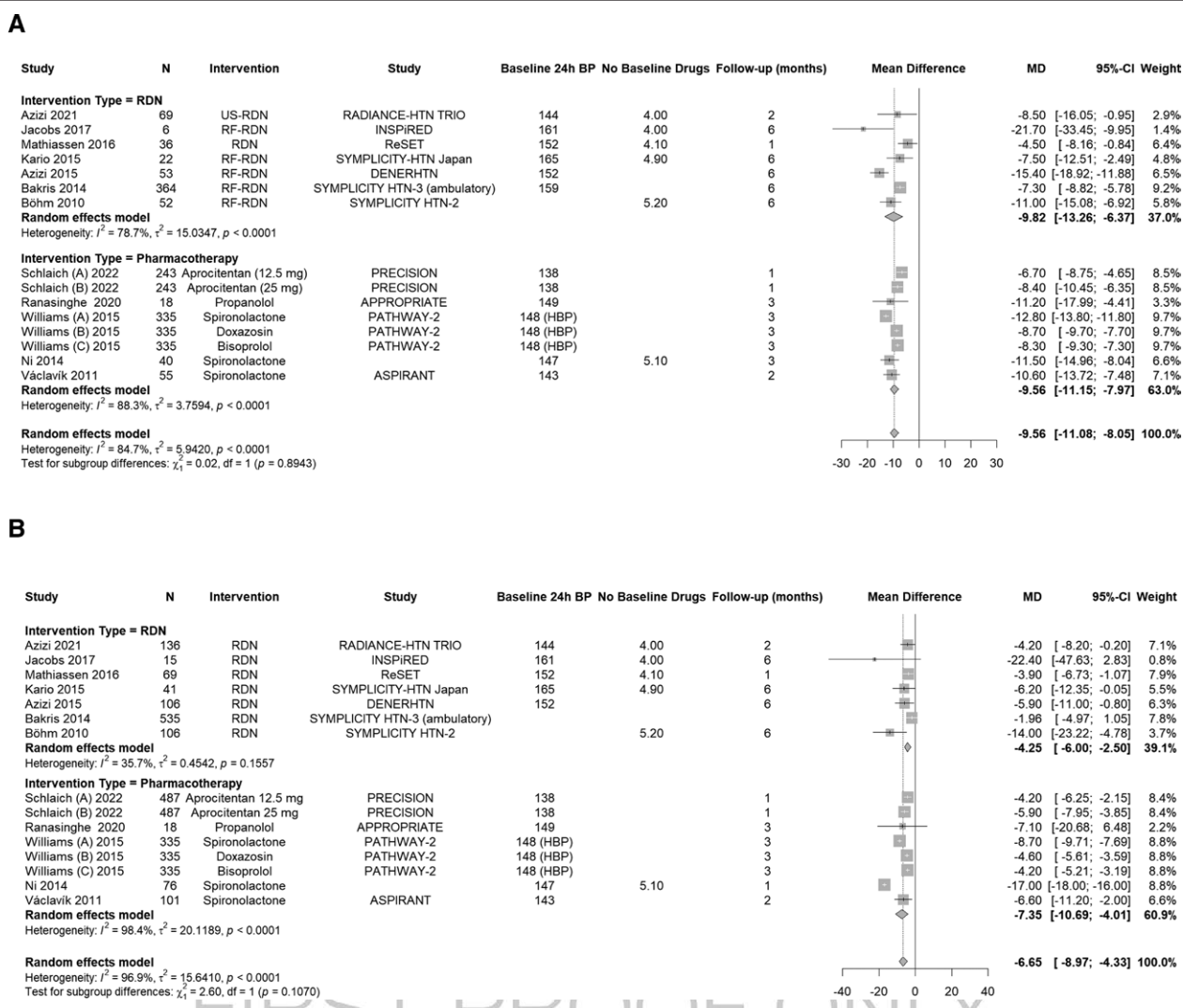


Figure 4. Meta-analysis of mean 24-hour ambulatory systolic blood pressure (BP) difference in placebo/sham-controlled randomized studies: (A) 12 studies, 15 reports, n=2206 participants, 27% renal denervation [RDN], analyzed as single-arm studies; and (B) 12 studies, 15 reports, n=3182 participants, 32% RDN, analyzed in the context of placebo/sham-corrected analyses.

RF indicates radiofrequency.

interventions, reflecting the greater number but smaller sample size of RDN studies.

Among pharmacological interventions, mineralocorticoid receptor antagonists, particularly spironolactone, were the most frequently studied, followed by β -blockers, α -blockers, and centrally acting agents, in line with guideline recommendations.¹⁻⁴ Drug dosing and treatment timing were not standardized across studies. Medication adherence was rarely verified using biological testing, despite patients receiving a mean of 4.8 antihypertensive agents at baseline. Some of the recent trials of novel agents (eg, baxdrostat, lorundrostat, and zilbesiran) were excluded because of methodological limitations (eg, absence of ambulatory BP data) or inclusion of mixed patient populations (eg, inclusion of both uncontrolled and resistant hypertension patients);

interestingly, the achieved average BP reductions were in the same levels as in the included studies (eg, average systolic office BP reduction of about 17 mmHg in the active treatment arms of the BaxHTN [baxdrostat] and Launch-HTN [lorundrostat] randomized clinical trials).⁷⁸⁻⁸⁰ Importantly, other landmark studies such as the PRECISION trial (aprocitentan) were deemed eligible for inclusion.¹⁴

Overall, RDN achieved systolic BP reductions comparable to those observed with pharmacotherapy. Previous studies in this field have produced inconsistent results, likely due to differences in study selection and analytical methods.^{81,82} The present meta-analysis combined randomized and nonrandomized studies using single-arm analyses to reduce heterogeneity and enhance comparability. Sensitivity analyses limited to placebo- or

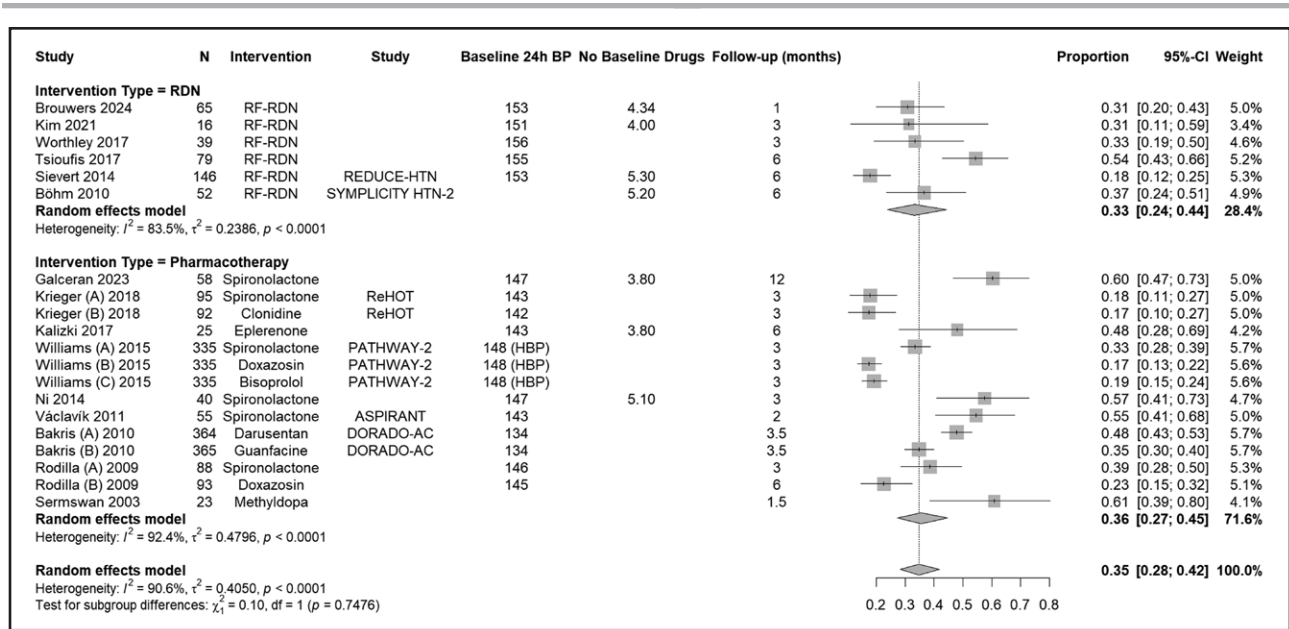


Figure 5. Hypertension control rate after the intervention employed in each study. The forest-plot is a stratified analysis comparing blood pressure (BP) reduction in pharmacotherapy vs renal denervation studies (nonsignificant difference). RDN indicates renal denervation; and RF, radiofrequency.

sham-controlled trials (probably the most important sensitivity analysis) demonstrated smaller BP reductions; however, nonsignificant difference between pharmacotherapy versus RDN studies. Notably, inference based on single-arm analyses may reflect a mixture of true treatment effects and nonspecific within-patient changes; however, it is not totally unjustified because these may reflect the real-world everyday clinical practice BP reductions. Other sensitivity analyses taking into consideration the sample size, the follow-up duration, and the origin of the included studies indicated a modest heterogeneity in BP reduction among studies conducted in different geographic regions. Although most studies had a low overall RoB, some were downgraded due to small sample sizes, unclear BP measurement protocols, or nonconsecutive patient recruitment. Meta-regression analyses indicated that greater BP reductions were associated with higher baseline BP and fewer concomitant medications, possibly reflecting regression to the mean, disease severity, or suboptimal medication adherence.

Traditionally, the primary goal in RHTN management has been the effective BP reduction, irrespective of the therapeutic modality. Notably, the cornerstone of hypertension management is the implementation of appropriate lifestyle interventions such as body weight reduction. This is of high importance given that the average body mass index of the included studies was 32 kg/m² and considering the availability of modern therapies for obesity including glucagon-like peptide-1 receptor agonists and tirzepatide. As far as the pharmacotherapy interventions are concerned, the PATHWAY-2 trial first established spironolactone as the most effective fourth-line agent for patients already receiving triple

therapy, outperforming bisoprolol and doxazosin.⁴⁸ Subsequently, randomized sham-controlled trials in carefully selected patients with true RHTN confirmed that RDN significantly reduces both office and ambulatory BP, with effects sustained for up to 9 years in real-world registries.^{83,84} Consequently, current ESH guidelines recommend adding spironolactone to standard triple single-pill combination (SPC) therapy or considering RDN as an adjunctive option, while ESC and ACC guidelines recommend spironolactone first, followed by RDN when pharmacological therapy is contraindicated or poorly tolerated.^{1–3} Notably, none of these recommendations are based on evidence for hard cardiovascular outcomes. Despite the above stepwise treatment algorithm for RHTN, our meta-analysis shows that BP control remains suboptimal: only one-third of patients with RHTN achieved target BP following a single additional intervention beyond the standard triple SPC. Interestingly, hypertension control definition in the majority of studies was based on the less intensive BP target of office BP <140/90 mmHg, and according to recent guidelines recommending a target BP of <130/80 mmHg, control rates would be expected to be even lower. Control rates were even lower among patients with established cardiovascular disease, underscoring the challenges of managing high-risk populations. These findings suggest that RHTN may be better conceptualized as difficult-to-control hypertension, highlighting the need for combined and complementary treatment strategies rather than reliance on a single additional intervention beyond the standard triple SPC therapy. The present review supports guideline recommendations advocating a complementary role for RDN alongside

pharmacological therapy. Integrating RDN with optimized pharmacotherapy beyond the triple SPC therapy may maximize BP reduction, reduce additional drug burden, and improve adherence through the sustained, always-on effects of RDN. Accordingly, we propose that a treatment strategy that incorporates RDN as part of a comprehensive, individualized approach, alongside tailored selection of a fourth-line agent (eg, mineralocorticoid receptor antagonist, β -blocker or novel agents, such as aldosterone synthase inhibitors or endothelin receptor antagonists) based on patient phenotype and comorbidities should be evaluated in future clinical trials (Graphic Abstract). This approach represents a shift in RHTN management—from escalation of stepwise multiple medications toward personalized, combination-based, and durable therapeutic strategies. In these lines, emerging quadruple SPC regimens are currently under investigation and may play an important role in future treatment pathways for RHTN.⁸⁵ These considerations warrant evaluation in future clinical trials.

The findings of this systematic review and meta-analysis should be interpreted in light of several limitations. First, there was substantial interstudy heterogeneity in factors such as patient characteristics, BP measurement protocols, drug dosages, hypertension control definitions, and RDN implementation methods, which, however offered the opportunity for performing sensitivity and meta-regression analyses. Second, most studies had small sample sizes, reflecting the limited amount of high-quality research in this area. Nevertheless, sensitivity analyses were performed to account for small sample sizes and short follow-up periods based on thresholds derived by distribution criteria from the included studies. Third, some studies—particularly long-term RDN trials not primarily designed for this research question—did not fully report baseline treatment changes, although the most recent follow-up data were analyzed to minimize bias. Fourth, the use of a single-arm analyses enhanced comparability across studies but also required sensitivity analyses to address potential placebo or sham effects. Fifth, certain populations were overrepresented, particularly in crossover studies in which the same participants received multiple treatment regimens. Finally, although inclusion criteria ensured that only patients with true RHTN were analyzed (confirmed through out-of-office BP assessment and exclusion of secondary hypertension), medication adherence was poorly documented in most studies. Future well-designed trials incorporating novel pharmacological agents and device-based therapies are needed to better define optimal treatment strategies for patients with RHTN.

PERSPECTIVES

This systematic review and meta-analysis demonstrates that adding a single intervention—either pharmacological

therapy or RDN—to guideline-directed background treatment with a triple SPC is insufficient to achieve BP control in most patients with RHTN. Overall, only approximately one-third of patients attain adequate BP control with this approach. These findings underscore the need for future research and clinical practice to move beyond single add-on therapies and adopt combination treatment strategies that integrate optimized pharmacological regimens beyond the standard triple SPC with device-based interventions. Such an integrated approach should be evaluated in future clinical trials and may offer more effective and durable BP control in this particularly challenging patient population.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S9

Figures S1–S18

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